

The background of the slide is a photograph of Mount Fuji, a snow-capped mountain, rising behind a town and a body of water. The mountain's peak is covered in white snow, and its slopes are a mix of brown and white. The town below the mountain is visible, with several buildings and trees. The water in the foreground is calm, reflecting the mountain and the sky. The sky is a clear, bright blue.

WAVEFORM TUNING – FONTYS PRJ4

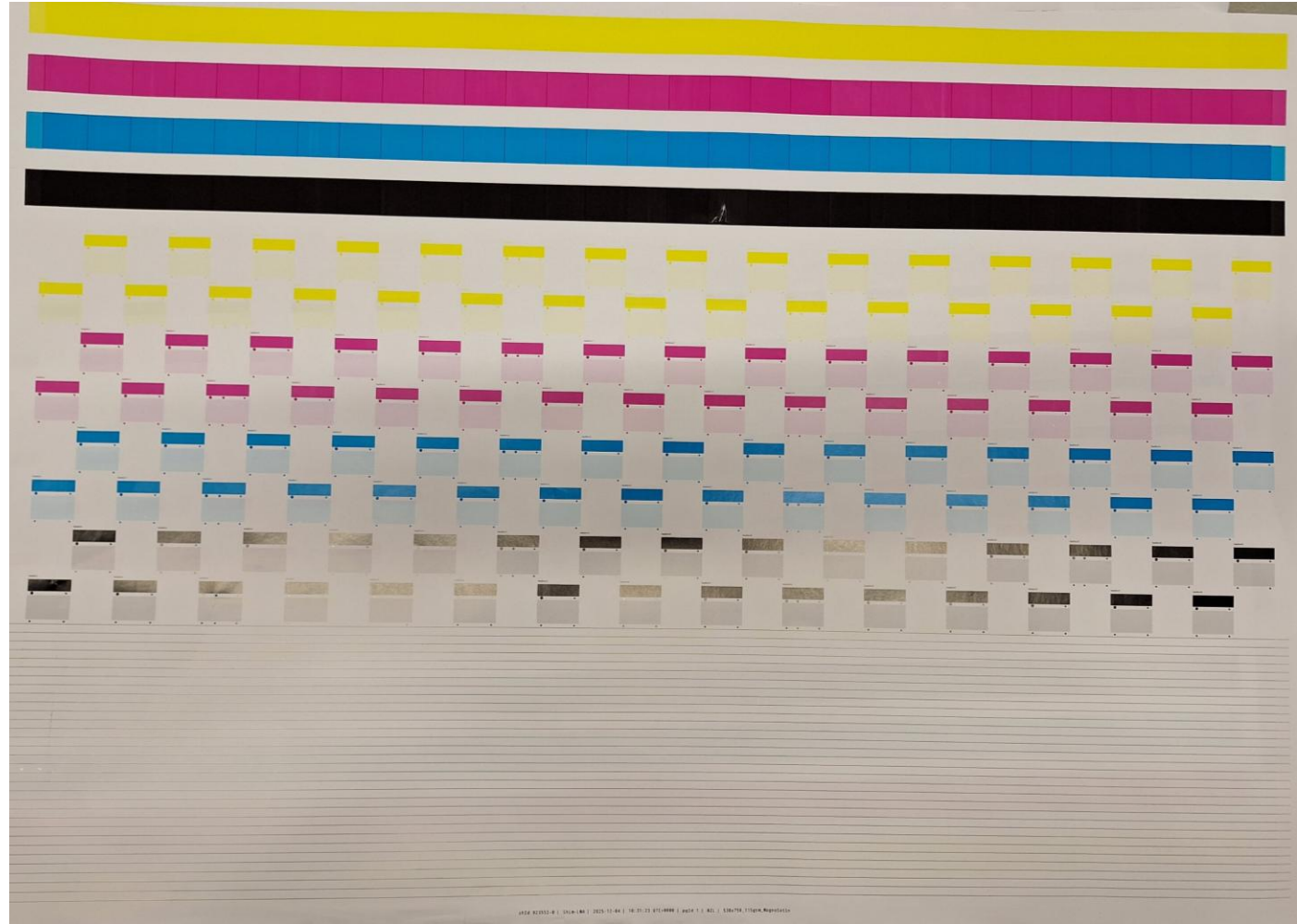
INTRODUCTORY MEETING

STEEF REIJNTJES, TIM VAN ZOGGEL

5-3-2026

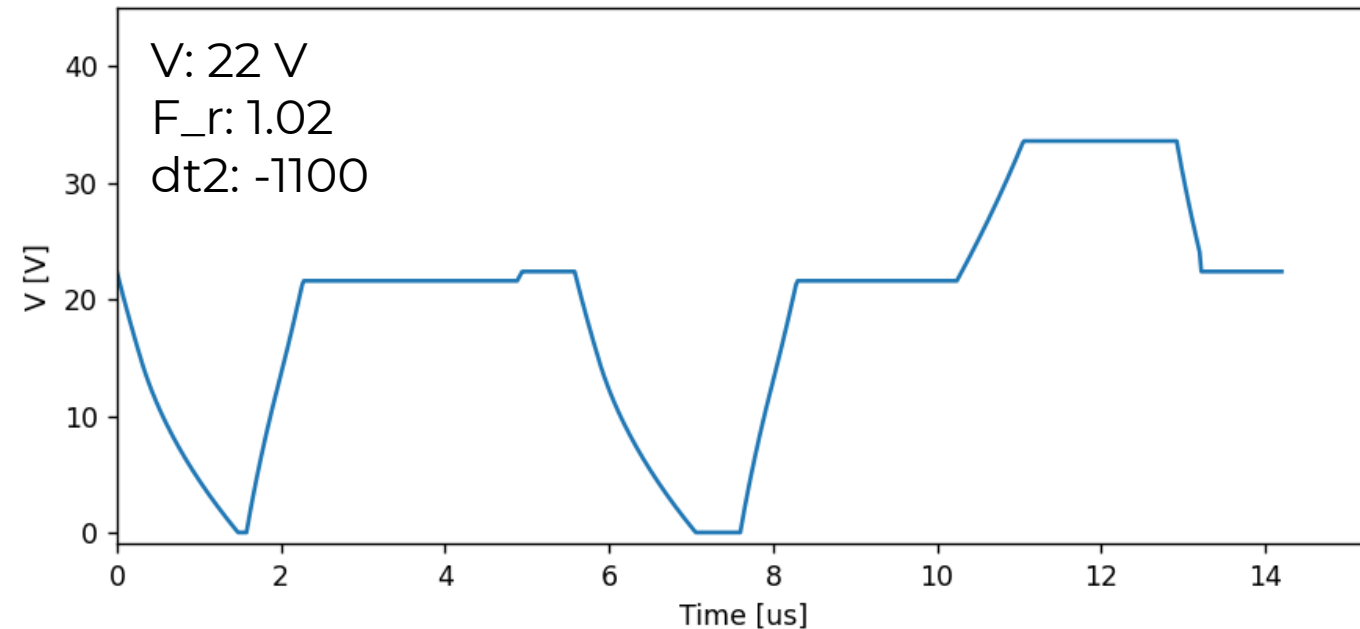
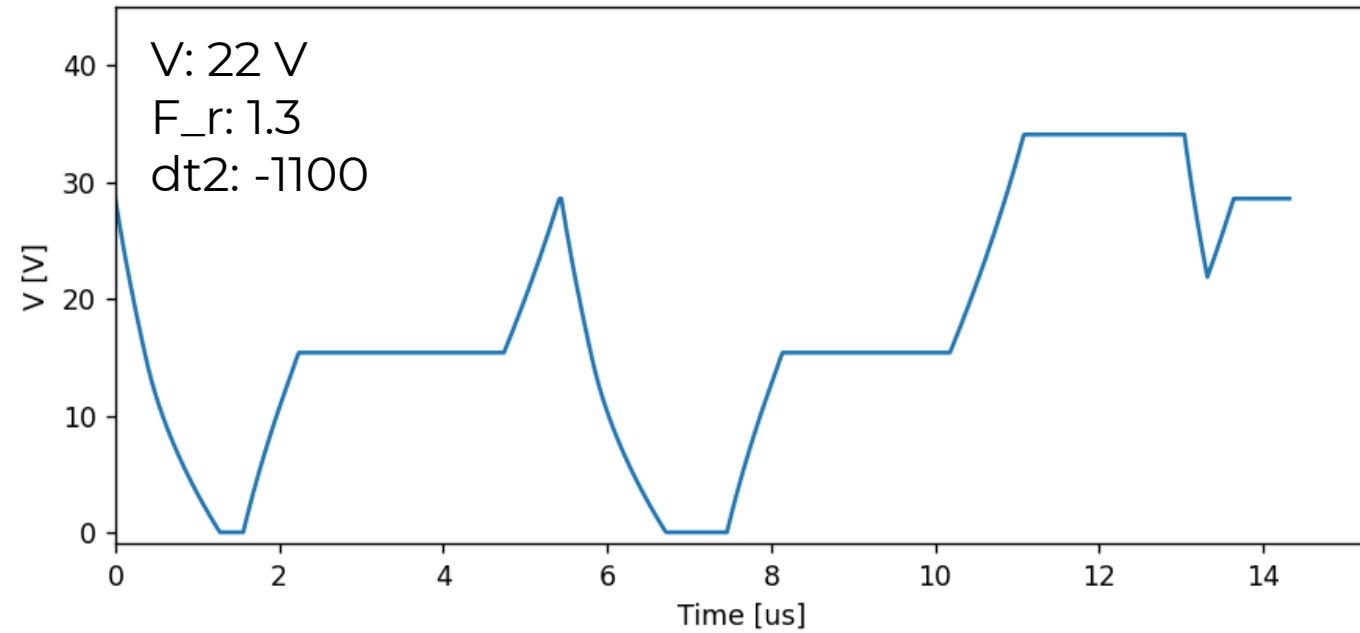
PRINTER DETAILS

- A printer has 8 print heads for every color in [C, M, Y, K]
 - Every print head has 4 chips.
- 32 chips in total, but we skip 0 and 31, leaving us with 30 individual chips.
- Every chip has 1136 nozzles, with the final 16 nozzles overlapping, so 1120 nozzles used



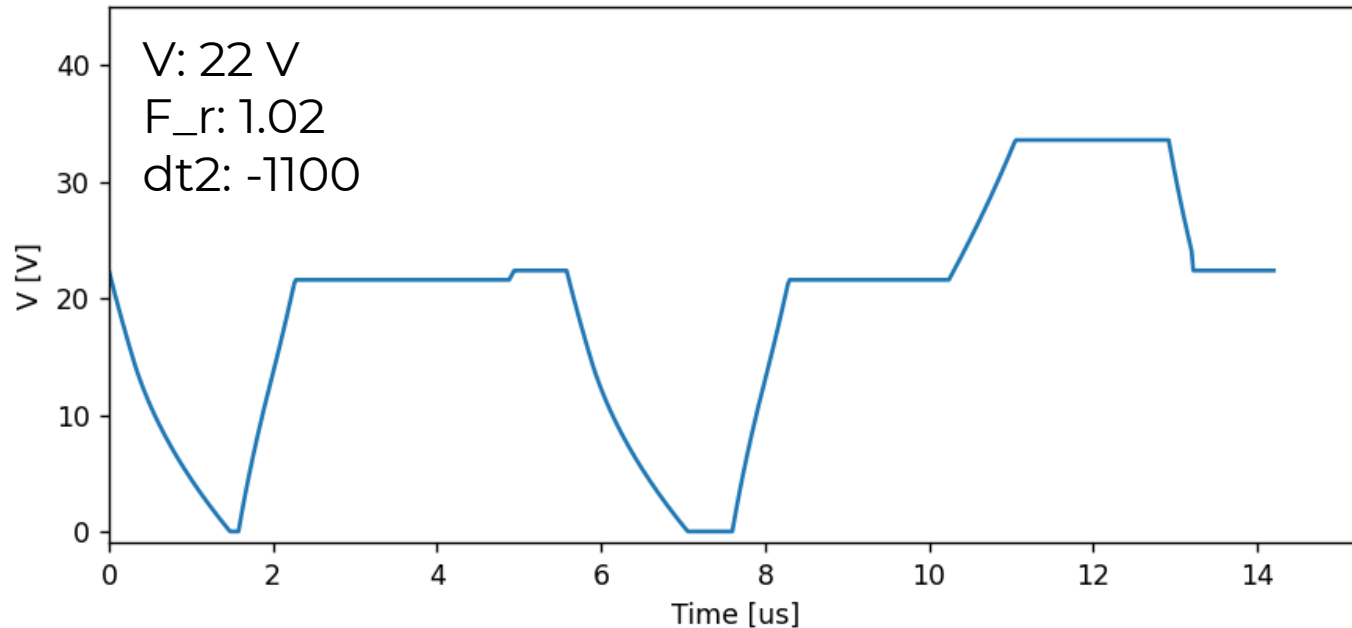
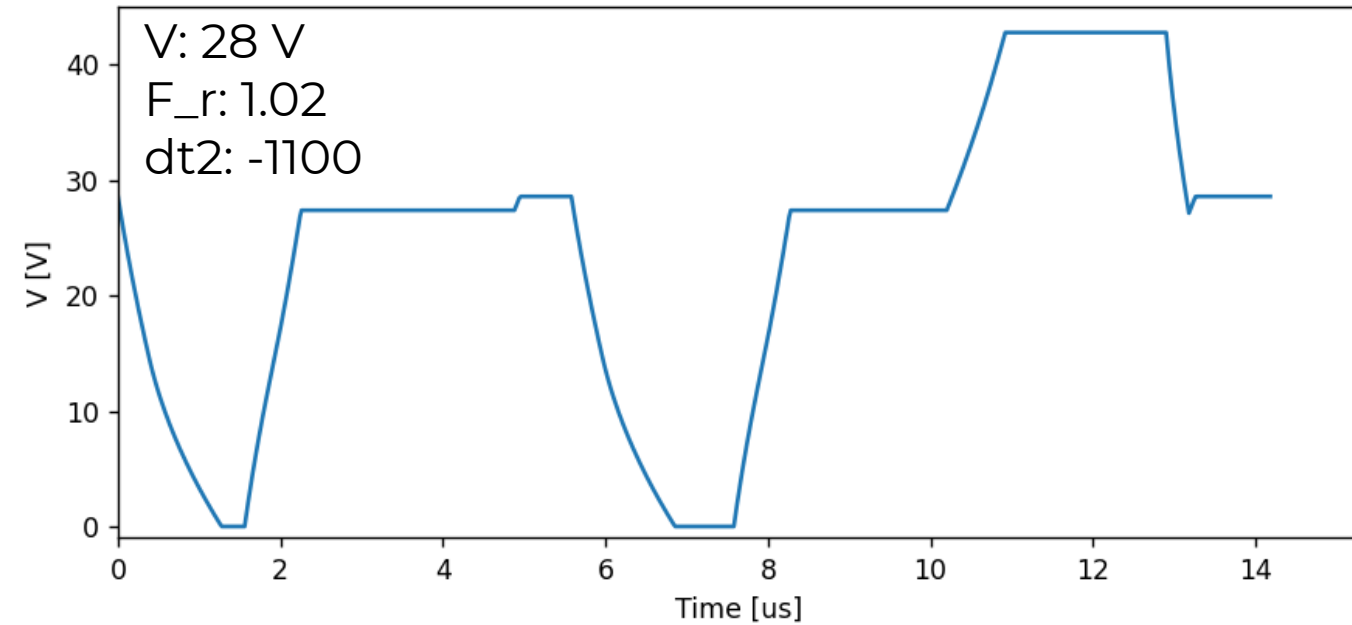
WAVEFORM DETAILS

- 3 parameters to tune:
 - Voltage
 - Flank ratio
 - Delta t2 (dt2)
- Voltage and Flank ratio are *per chip* parameters
- dt2 is a per nozzle parameter



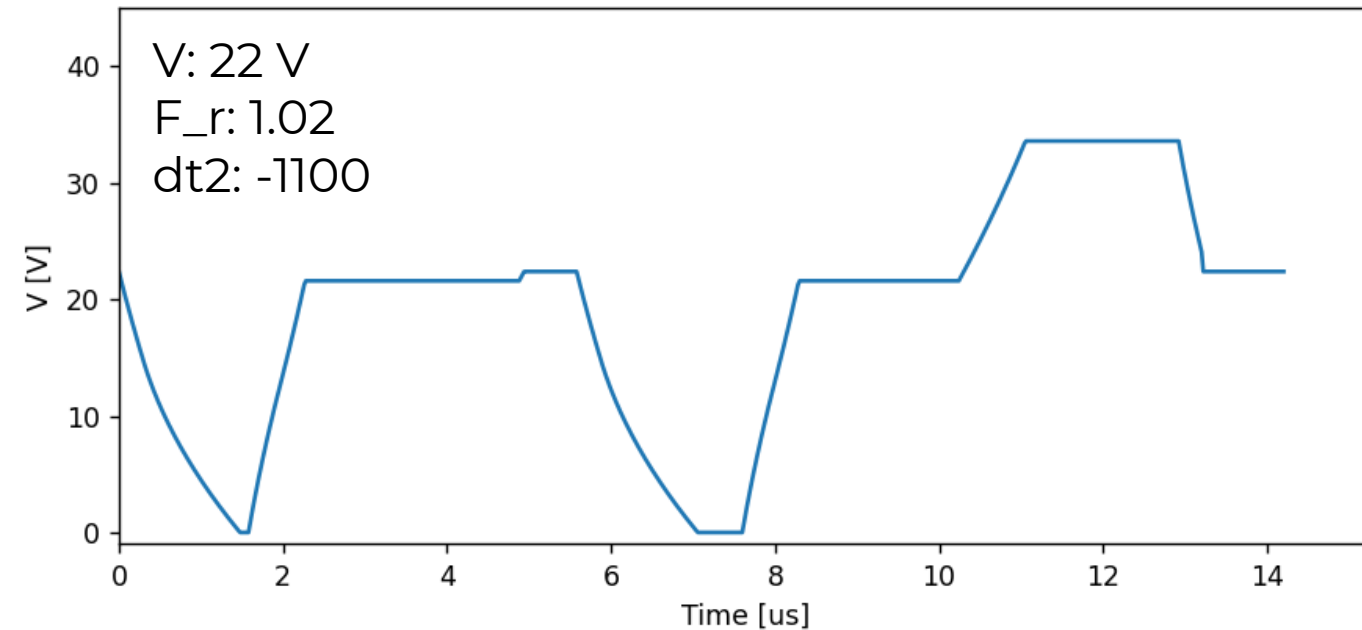
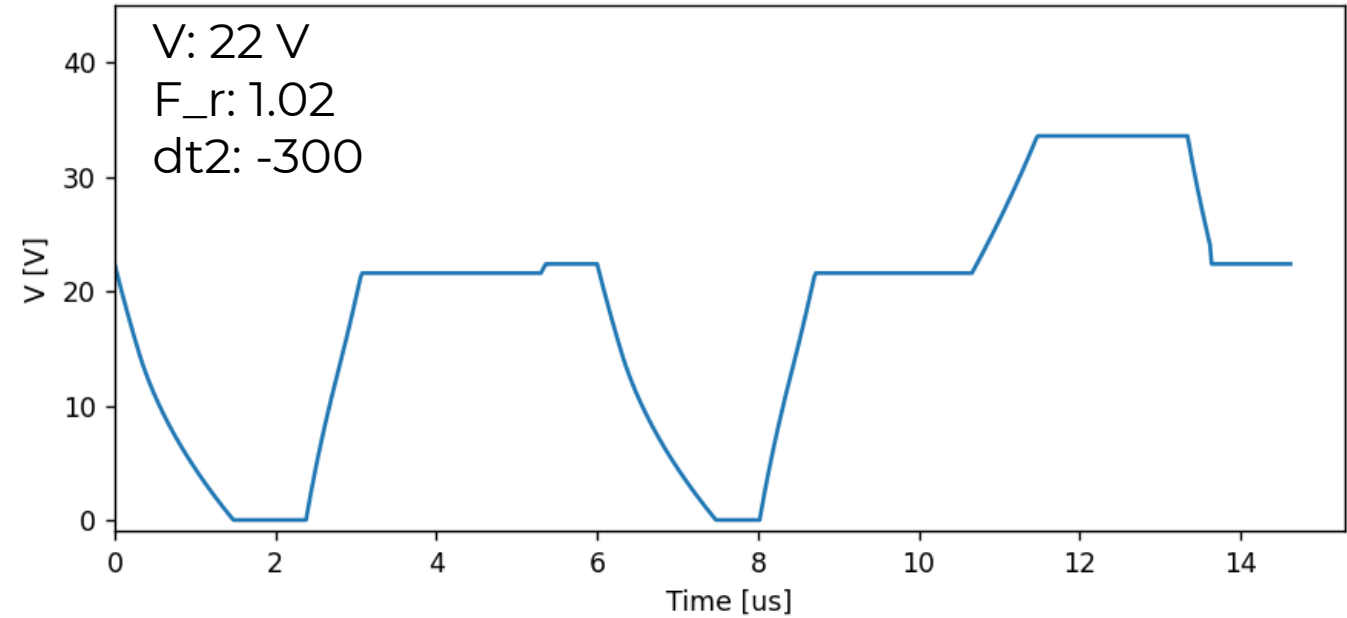
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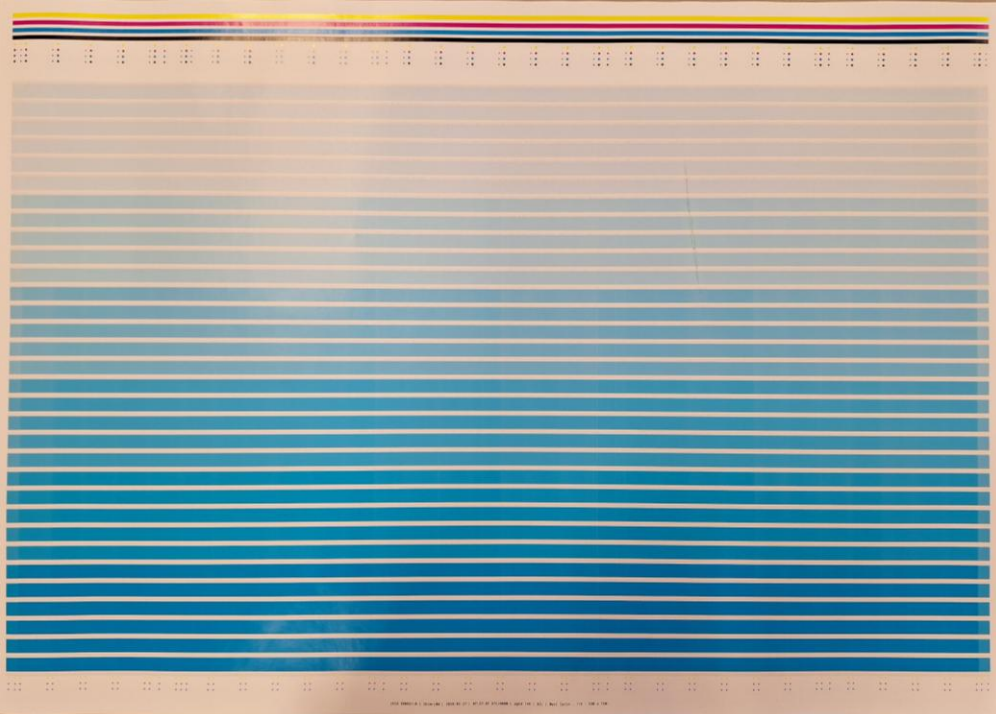
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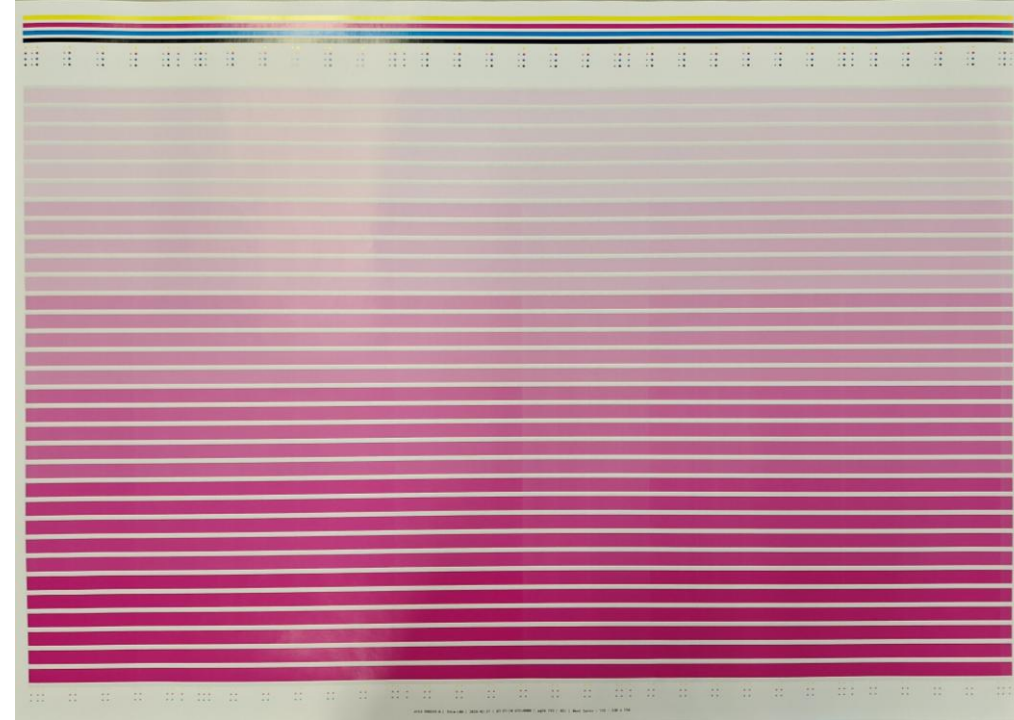
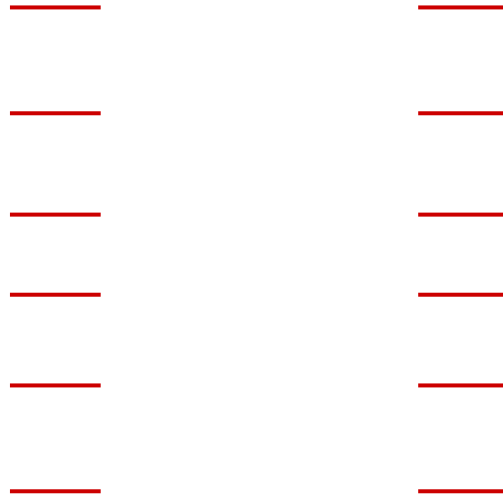


WHAT TYPE OF PRINTS ARE USED TO GET DATA

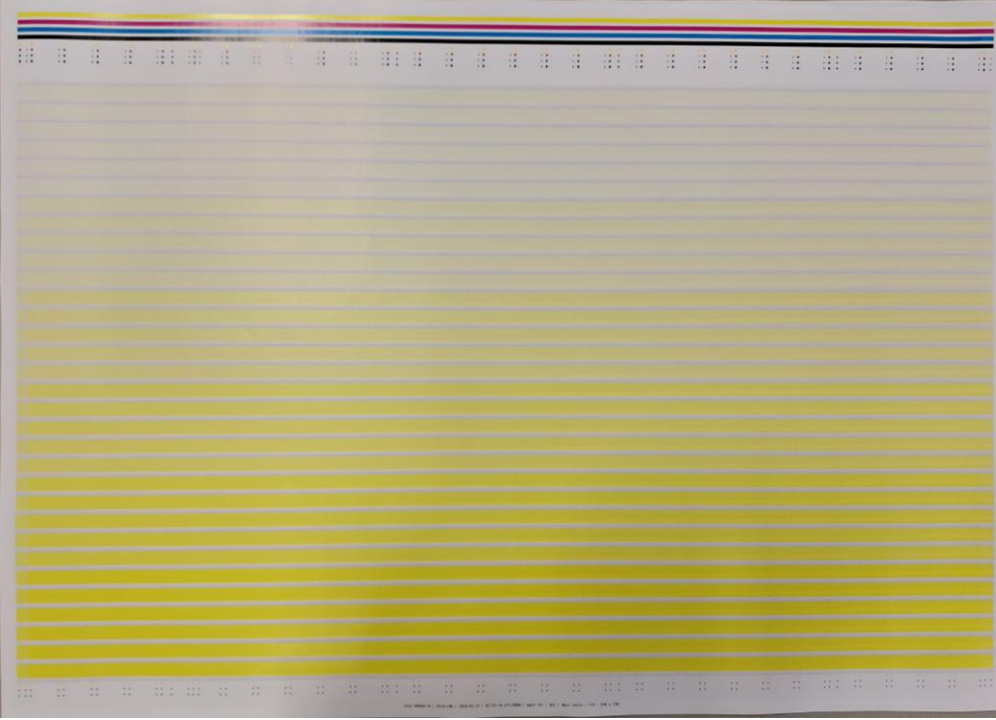
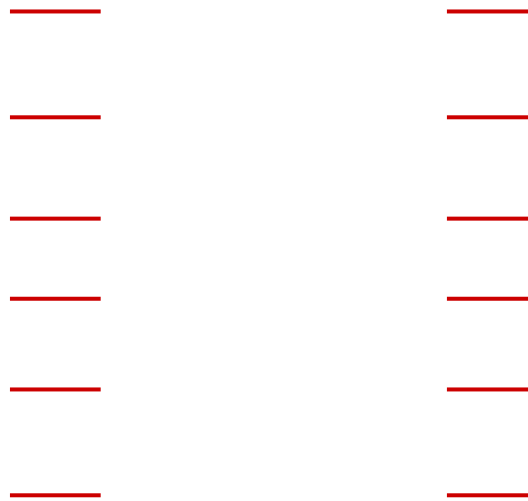
- Up front, a Voltage and Flank ratio were defined, e.g. {22, 1.02}, {24, 1.28} etc.
- 5 dt2 are defined up front: [-1100, -900, -700, -500, -300]
- For a set of prints, every chip for every color is given the same {V, F_r}
- On every sheet 6 'coverages' (amount of ink) are printed
 - Every coverage is printed 5 times, once using a different dt2 from the above list for every single nozzle
- This means every sheet has 30 'sets' of data per chip for a given {V, F_r}
- Every sheet is printed 6 times with the exact same settings



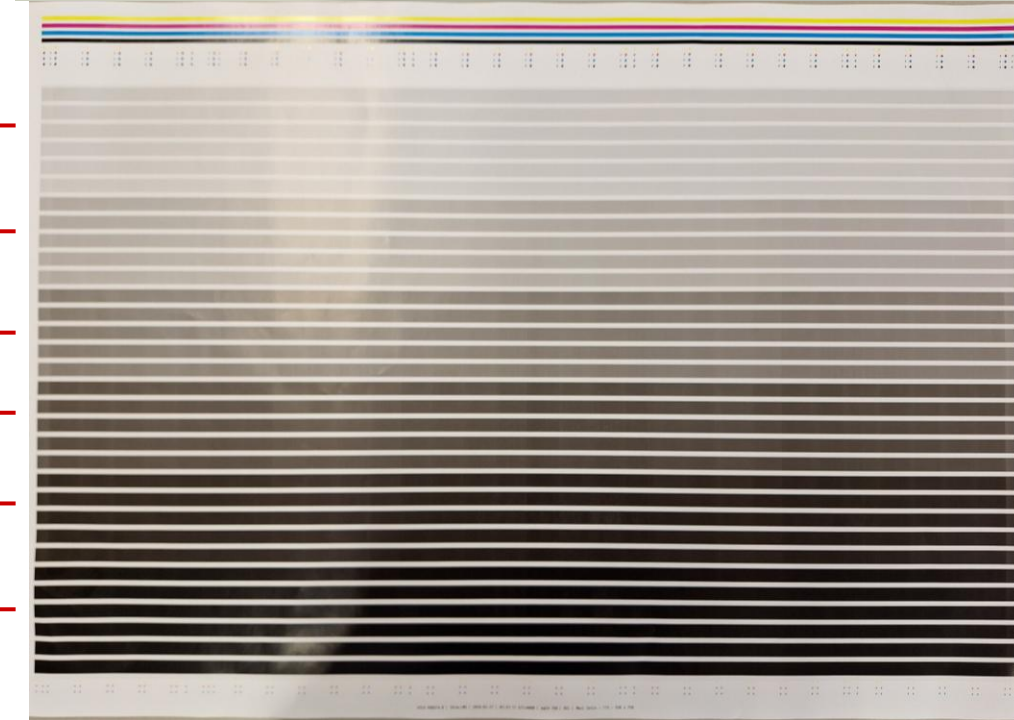
6 coverages
per color



From little ink to a
large amount of ink



5 dt2 per coverage



WHAT IS BEING MEASURED

- When such a test chart is printed:
- The scanner scans the sheet, processing is done:
 - For every nozzle (1120), in every chip (30)
- A Scanner Density (SD) value is logged, for every single coverage (also 30).
- This means 1 sheet gives us $1120 * 30 * 30 = 1,008,000$ SD values
- Scanner Density tells us something about how dark a color is:
 - 0 means there is no color at all
 - 1 means the darkest possible color

DATA SET

- 4 colors
 - 30 chips per color
 - 1120 nozzles per chip
 - 30 different {V, F_r} combinations
 - 5 dt2 per {V, F_r}
 - 6 coverage levels per {V, F_r, dt2}
 - 6 repeated sheets per waveform set
-
- Total of: $4 * 30 * 1120 * 30 * 5 * 6 * 6 = 725,760,000$ individual samples

THINGS TO KEEP IN MIND

- Different colors behave differently
- Every single chip behaves differently
 - Every nozzle in a chip behaves slightly differently
- Scanner resolution is lower than print resolution:
 - Ink from neighboring nozzles influences SD measurement of a nozzle

Canon